TONER-1: a frontier model with a paper trail.

State of the art among language models that run entirely inside a print job.

> The most capable open-weight, US-Letter-native, sub-300K-parameter, PostScript-native, zero-host-FLOP, duplex-ready language model running on hardware already in your copy room. *Evaluated on 20 lb bond at room temperature.*

SPECIFICATIONS

Base model	Karpathy's llama2.c stories260K (trained on TinyStories), fine-tuned by us on 8,757 printer stories written by Xing4.0-29B-A4B (Apache-2.0)
Parameters	260,032
Architecture	Llama-2 decoder: 5 layers, dim 64, 8 query heads / 4 KV heads, head size 8, SwiGLU FFN (hidden 172), RoPE, RMSNorm
Vocabulary	512-token SentencePiece
Quantization	int8 weights, one scale per row; activations in PostScript reals; embedding shared with the output head
Context window	8.5 × 11 in. (512 tokens)
KV cache	Persistent, archival, recyclable
Job size	595 KB, one file: `toner1.ps`. Weights, vocabulary, a forward pass of about 100 lines and its own page layout
Speed, measured	About 80 tok/s in Ghostscript 10.07 on an Apple-silicon Mac, using 3.3 to 7.5 MB of PostScript VM
Speed, printer	0.92 tok/s, measured on an HP Color LaserJet Pro MFP M281fdw: 200 tokens in 3:37 by the printer's own clock
Time to first token	1.7 s on the HP, read back over the network
Accelerator	The office's hottest inference chip (fuser, typically about 200 °C)
Host compute	0 FLOPs. The host only sends bytes: `nc <printer-ip> 9100 < toner1.ps`
Output modality	Native toner
Open weights	Literally. Hex appendix available; OCR to load
Pricing	Inference cost includes toner

INTENDED USE

Writing short children's stories, mostly about printers, one US Letter sheet at a time, on a printer you own or have permission to use. Demonstrating that a print job is a program. *Out of scope: prompts. The model chooses the plot.*

LIMITATIONS

- It is 260K parameters.
- It mostly writes about Lily, Tim and the printer.
- It will hallucinate a quiet salad.
- No prompt input: every story starts from the beginning-of-sequence token.
- Needs a PostScript interpreter, Level 2 or later. PCL- or GDI-only printers may print the file as text.
- Long jobs may hit the printer's job timeout or run out of VM.

SAFETY

- Refuses to generate while Tray 2 is open. Close the tray to proceed.
- **Air-gapped mode:** unplug Ethernet once the job has uploaded.
- **Kill switch:** the power button.
- Outputs are fused to paper and can no longer execute.

PRODUCT LINEUP

TONER-1 — AVAILABLE

260K, PRINTER-TUNED / 595 KB JOB

The everyday copy-room model, now fine-tuned on printer stories. Under 4 s per page in Ghostscript; 3:37 for 200 tokens on an HP printer.

TONER-1 Pro — EST. 20 MIN–5 H

12.2M, OURS / 24.9 MB JOB

Trained from scratch by us, 47 times the parameters. Estimated: send it after lunch, collect your story by close of business.

TONER-1 Classic — PLANNED

1985 LASERWRITER / 68000

A 1985-native AI PC. The LaserWriter's 12 MHz 68000 was faster than the 7.8 MHz Macintosh it was plugged into.

TONER-1 is a working print job and a parody of frontier-model launches. This handout is a PostScript program, too.
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Undefeated inside a print job.

TONER-1 posts a result on every benchmark it can enter. Frontier models from major labs were offered identical conditions: one print job, one sheet, zero host FLOPs.

FIGURE 1 HEADLINE RESULTS (PARODY)

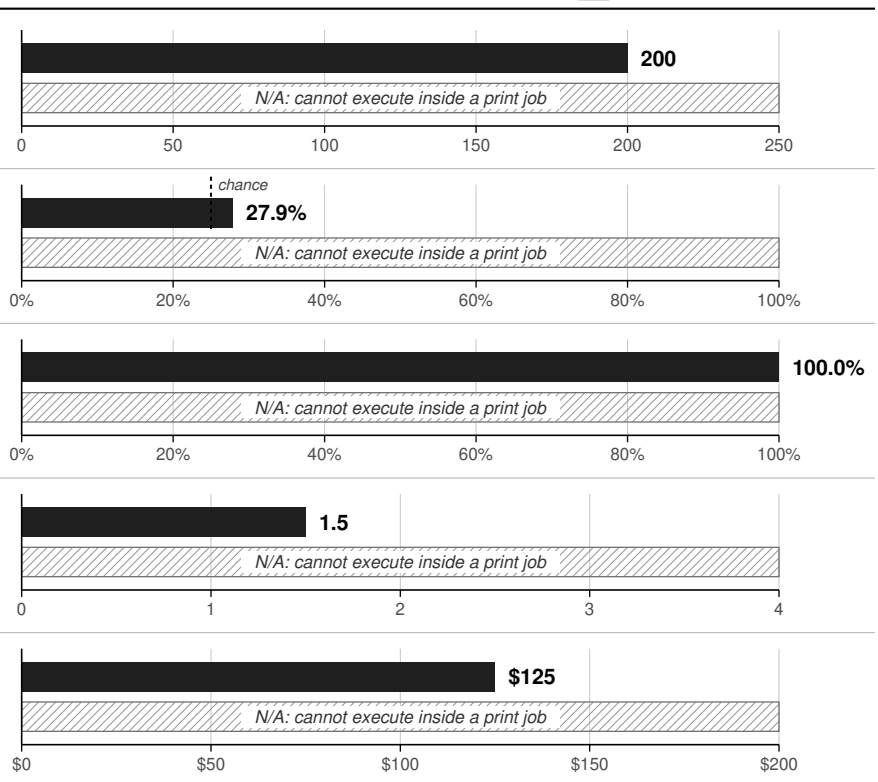
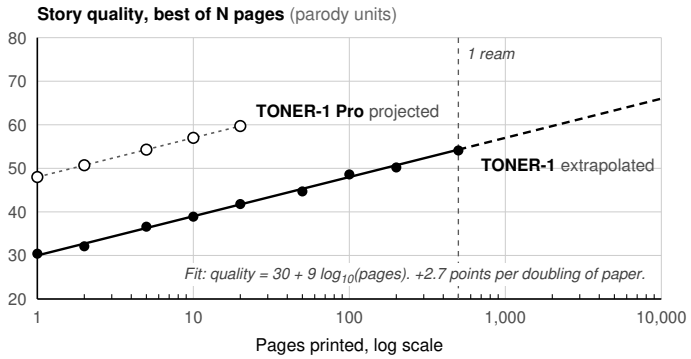


FIGURE 2 TEST-TIME SCALING



EVALUATION SETUP

Interpreter	Ghostscript 10.07, Apple-silicon Mac; HP PostScript 3017
Printer	HP Color LaserJet Pro MFP M281fdw
Speed	Measured in Ghostscript and on the printer
Paper	US Letter, simplex
Sampling	Temperature 0.8, seed 42, 200 steps
Prompt	None. Every run starts from BOS.
Ambient	Room temp.; fuser typically about 200 °C
Frontier models	Not run: no print-job support

REPRODUCE

```
nc <printer-ip> 9100 < toner1.ps
```

Parody benchmarks. Not real evaluations. TONER-1 speed measured in Ghostscript and on one printer. Frontier models from major labs were not run on these benchmarks and are not named; N/A means only that they cannot execute inside a print job. Figure 2 is parody data. TONER-1 is Karpathy's llama2.c stories260K, fine-tuned by us on printer stories, exported to int8 and run by a PostScript forward pass. TONER-1 Pro is a 12.2M-parameter model we trained from scratch, with our own 4,096-token tokenizer. Prior art: int-llm-postscript (Aug 2026) ran a 14K-parameter microGPT in PostScript under Ghostscript, and PSChess played chess on real printers. On 2026-11-05 TONER-1 ran on an actual printer, an HP Color LaserJet Pro MFP M281fdw. Only use printers you own or have permission to use.

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